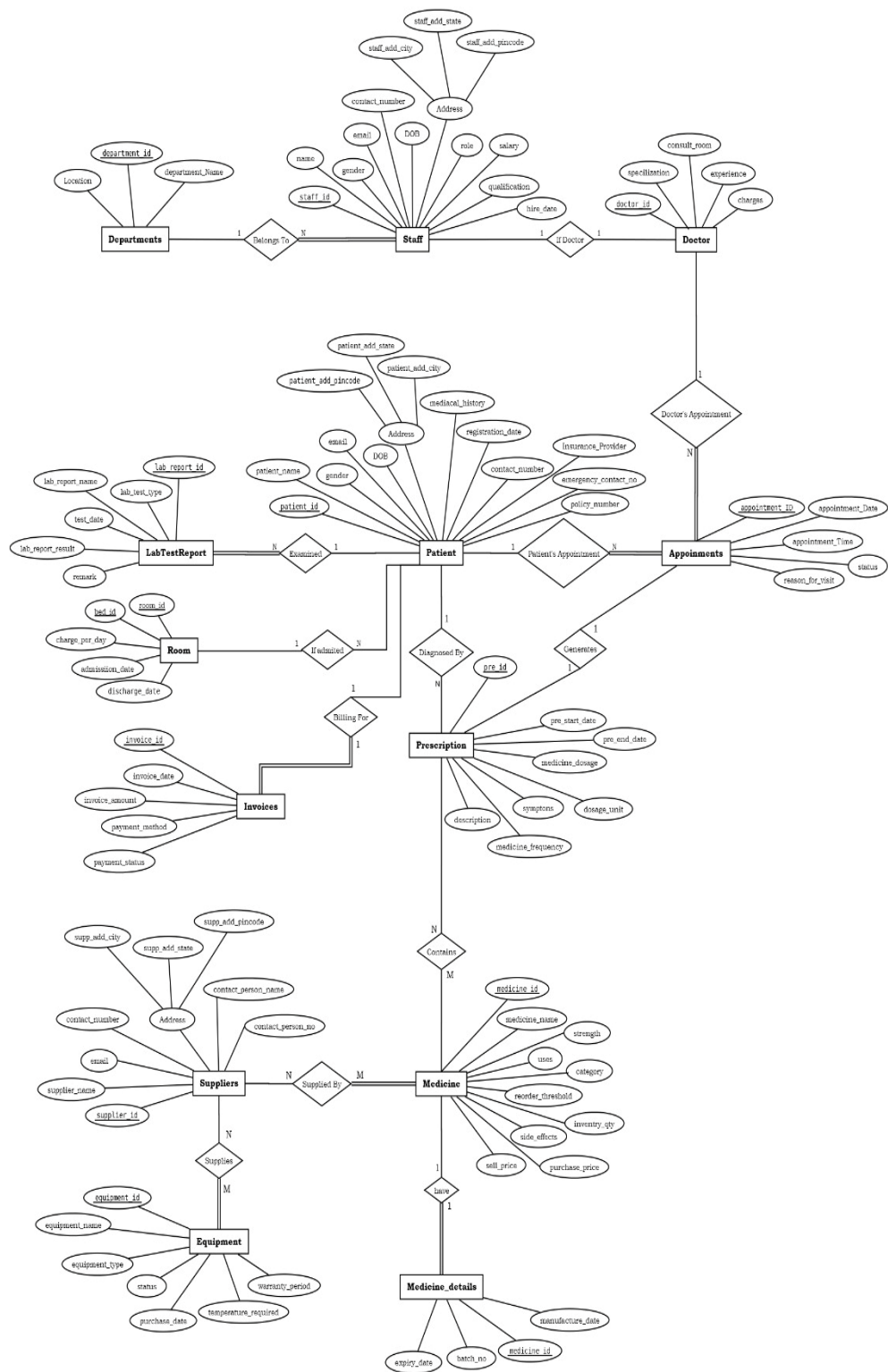
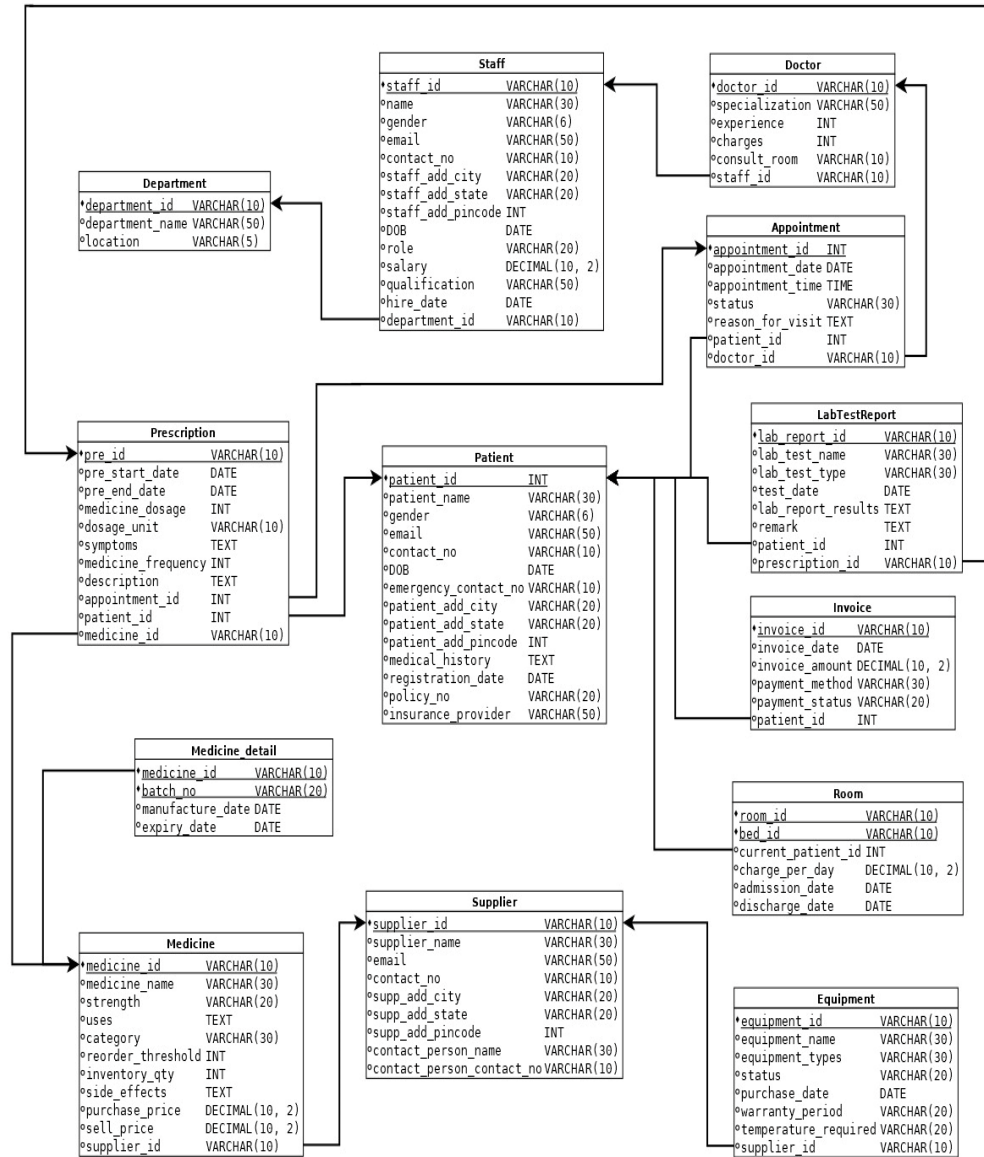


ER DIAGRAM



RELATIONAL SCHEMA



Normalization Proofs

1. Department Table

Attributes: department_id, department_name, location

Candidate Key: {department_id}

Functional Dependencies:

department_id → department_name, location

2. Staff Table

Attributes: staff_id, name, gender, email, contact_no, staff_add_city, staff_add_state, staff_add_pincode, DOB, role, salary, qualification, hire_date, department_id

Candidate Key: {staff_id}

Functional Dependencies:

staff_id → name, gender, email, contact_no, staff_add_city, staff_add_state, staff_add_pincode, DOB, role, salary, qualification, hire_date, department_id

3. Doctor Table

Attributes: doctor_id, specialization, experience, charges, consult_room, staff_id

Candidate Key: {doctor_id}

Functional Dependencies:

doctor_id → specialization, experience, charges, consult_room, staff_id

4. Supplier Table

Attributes: supplier_id, supplier_name, email, contact_no, supp_add_city, supp_add_state, supp_add_pincode, contact_person_name, contact_person_contact_no

Candidate Key: {supplier_id}

Functional Dependencies:

supplier_id → supplier_name, email, contact_no, supp_add_city, supp_add_state, supp_add_pincode, contact_person_name, contact_person_contact_no

5. Medicine Table

Attributes: medicine_id, medicine_name, strength, uses, category, reorder_threshold, inventory_qty, side_effects, purchase_price, sell_price, supplier_id

Candidate Key: {medicine_id}

Functional Dependencies:

medicine_id → medicine_name, strength, uses, category, reorder_threshold, inventory_qty, side_effects, purchase_price, sell_price, supplier_id

6. Medicine_detail Table

Attributes: medicine_id, batch_no, manufacture_date, expiry_date

Candidate Key: {medicine_id, batch_no}

Functional Dependencies:

{medicine_id, batch_no} → manufacture_date, expiry_date

7. Equipment Table

Attributes: equipment_id, equipment_name, equipment_types, status, purchase_date, warranty_period, temperature_required, supplier_id

Candidate Key: {equipment_id}

Functional Dependencies:

equipment_id → equipment_name, equipment_types, status, purchase_date, warranty_period, temperature_required, supplier_id

8. Patient Table

Attributes: patient_id, patient_name, gender, email, contact_no, DOB, emergency_contact_no, patient_add_city, patient_add_state, patient_add_pincode, medical_history, registration_date, policy_no, insurance_provider

Candidate Key: {patient_id}

Functional Dependencies:

patient_id → patient_name, gender, email, contact_no, DOB, emergency_contact_no, patient_add_city, patient_add_state, patient_add_pincode, medical_history, registration_date, policy_no, insurance_provider

9. Appointment Table

Attributes: appointment_id, appointment_date, appointment_time, status, reason_for_visit, patient_id, doctor_id

Candidate Key: {appointment_id}

Functional Dependencies:

appointment_id → appointment_date, appointment_time, status, reason_for_visit, patient_id, doctor_id

10. Prescription Table

Attributes: pre_id, pre_start_date, pre_end_date, medicine_dosage, dosage_unit, symptoms, medicine_frequency, description, patient_id, medicine_id

Candidate Key: {pre_id}

Functional Dependencies:

pre_id → pre_start_date, pre_end_date, medicine_dosage, dosage_unit, symptoms, medicine_frequency, description, patient_id, medicine_id

11. LabTestReport Table

Attributes: lab_report_id, lab_test_name, lab_test_type, test_date, lab_report_results, remark, patient_id

Candidate Key: {lab_report_id}

Functional Dependencies:

lab_report_id → lab_test_name, lab_test_type, test_date, lab_report_results, remark, patient_id

12. Room Table

Attributes: room_id, bed_id, current_patient_id, charge_per_day, admission_date, discharge_date

Candidate Key: {room_id, bed_id}

Functional Dependencies:

{room_id, bed_id} → current_patient_id, charge_per_day, admission_date, discharge_date

13. Invoice Table

Attributes: invoice_id, invoice_date, invoice_amount, payment_method, payment_status, patient_id

Candidate Key: {invoice_id}

Functional Dependencies:

invoice_id $\rightarrow$ invoice_date, invoice_amount, payment_method, payment_status, patient_id